

Given that the conditions in binary star systems should make planetary formation nearly impossible, it's not surprising that the existence of planets in such systems has lacked _____ explanation. Roman Rafikov and Kedron Silsbee shed light on the subject when they used modeling to determine a complex set of factors that could support planets' development.

Which choice completes the text with the most logical and precise word or phrase?

- A. a discernible
- B. a straightforward
- C. an inconclusive
- D. an unbiased

Text 1

Films and television shows commonly include a long list of credits naming the people involved in a production. Credit sequences may not be exciting, but they generally ensure that everyone's contributions are duly acknowledged. Because they are highly standardized, film and television credits are also valuable to anyone researching the careers of pioneering cast and crew members who have worked in the mediums.

Text 2

Video game scholars face a major challenge in the industry's failure to consistently credit the artists, designers, and other contributors involved in making video games. Without a reliable record of which people worked on which games, questions about the medium's development can be difficult to answer, and the accomplishments of all but its best-known innovators can be difficult to trace.

Based on the texts, how would the author of Text 1 most likely respond to the discussion in Text 2?

- A. By recommending that the scholars mentioned in Text 2 consider employing the methods regularly used by film and television researchers
- B. By pointing out that credits have a different intended purpose in film and television than in the medium addressed by the scholars mentioned in Text 2
- C. By suggesting that the scholars mentioned in Text 2 rely more heavily on credits as a source of information than film and television researchers do
- D. By observing that a widespread practice in film and television largely prevents the kind of problem faced by the scholars mentioned in Text 2

Text 1

Growth in the use of novel nanohybrids—materials created from the conjugation of multiple distinct nanomaterials, such as iron oxide and gold nanomaterials conjugated for use in magnetic imaging—has outpaced studies of nanohybrids’ environmental risks. Unfortunately, risk evaluations based on nanohybrids’ constituents are not reliable: conjugation may alter constituents’ physiochemical properties such that innocuous nanomaterials form a nanohybrid that is anything but.

Text 2

The potential for enhanced toxicity of nanohybrids relative to the toxicity of constituent nanomaterials has drawn deserved attention, but the effects of nanomaterial conjugation vary by case. For instance, it was recently shown that a nanohybrid of silicon dioxide and zinc oxide preserved the desired optical transparency of zinc oxide nanoparticles while mitigating the nanoparticles’ potential to damage DNA.

Based on the texts, how would the author of Text 2 most likely respond to the assertion in the underlined portion of Text 1?

- A. By concurring that the risk described in Text 1 should be evaluated but emphasizing that the risk is more than offset by the potential benefits of nanomaterial conjugation
- B. By arguing that the situation described in Text 1 may not be representative but conceding that the effects of nanomaterial conjugation are harder to predict than researchers had expected
- C. By denying that the circumstance described in Text 1 is likely to occur but acknowledging that many aspects of nanomaterial conjugation are still poorly understood
- D. By agreeing that the possibility described in Text 1 is a cause for concern but pointing out that nanomaterial conjugation does not inevitably produce that result

Businesses typically try to promote themselves, but research by Michael Luca, Abhishek Nagaraj, and Gauri Subramani suggests that some do not _____ their marketing. The researchers studied restaurants in Texas and found that those listed on a popular review website (which can be done at no cost) had revenue increases of 5 percent on average, but 33 percent of restaurants studied never listed on the site.

Which choice completes the text with the most logical and precise word or phrase?

- A. adorn
- B. optimize
- C. endorse
- D. counteract

Some scientists have suggested that mammals in the Mesozoic era were not a very _____ group, but paleontologist Zhe-Xi Luo's research suggests that early mammals living in the shadow of dinosaurs weren't all ground-dwelling insectivores. Fossils of various plant-eating mammals have been found in China, including species like *Vilevolodon diplomylos*, which Luo says could glide like a flying squirrel.

Which choice completes the text with the most logical and precise word or phrase?

- A. predatory
- B. obscure
- C. diverse
- D. localized

Containing over 160 billion base pairs of DNA resulting from polyploidy (whole genome duplication) and accumulation of noncoding DNA without significant deletion, the genome of *Tmesipteris oblongeolata* (New Caledonian fork fern) is the largest known among eukaryotes—and, given the high biological costs of extremely large genomes, is unlikely to be _____.

Which choice completes the text with the most logical and precise word or phrase?

- A. protracted
- B. superseded
- C. curtailed
- D. obviated

Scholarly discussions of gender in Shakespeare's comedies often celebrate the rebellion of the playwright's characters against the rigid expectations _____ by Elizabethan society. Most of the comedies end in marriage, with characters returning to their socially dictated gender roles after previously defying them, but there are some notable exceptions.

Which choice completes the text with the most logical and precise word or phrase?

- A. interjected
- B. committed
- C. illustrated
- D. prescribed

The following text is adapted from George Eliot's 1871–72 novel *Middlemarch*.

[Mr. Brooke] had travelled in his younger years, and was held in this part of the country to have contracted a too rambling habit of mind. Mr. Brooke's conclusions were as difficult to predict as the weather.

As used in the text, what does the word “contracted” most nearly mean?

- A. Restricted
- B. Described
- C. Developed
- D. Settled

Space scientists Anna-Lisa Paul, Stephen M. Elardo, and Robert Ferl planted seeds of *Arabidopsis thaliana* in samples of lunar regolith—the surface material of the Moon—and, serving as a control group, in terrestrial soil. They found that while all the seeds germinated, the roots of the regolith-grown plants were stunted compared with those in the control group. Moreover, unlike the plants in the control group, the regolith-grown plants exhibited red pigmentation, reduced leaf size, and inhibited growth rates—indicators of stress that were corroborated by postharvest molecular analysis.

Which choice best states the main purpose of the text?

- A. It describes an experiment that addressed an unresolved question about the extent to which lunar regolith resembles terrestrial soils.
- B. It compares two distinct methods of assessing indicators of stress in plants grown in a simulated lunar environment.
- C. It presents evidence in support of the hypothesis that seed germination in lunar habitats is an unattainable goal.
- D. It discusses the findings of a study that evaluated the effects of exposing a plant species to lunar soil conditions.

Studying late nineteenth- and early twentieth-century artifacts from an agricultural and domestic site in Texas, archaeologist Ayana O. Flewellen found that Black women employed as farm workers utilized hook-and-eye closures to fasten their clothes at the waist, giving themselves a silhouette similar to the one that was popular in contemporary fashion and typically achieved through more restrictive garments such as corsets. Flewellen argues that this sartorial practice shows that these women balanced hegemonic ideals of femininity with the requirements of their physically demanding occupation.

Which choice best states the main purpose of the text?

- A. To describe an unexpected discovery that altered a researcher's view of how rapidly fashions among Black female farmworkers in late nineteenth- and early twentieth-century Texas changed during the period
- B. To discuss research that investigated the ways in which Black female farmworkers in late nineteenth- and early twentieth-century Texas used fashion practices to resist traditional gender ideals
- C. To evaluate a scholarly work that offers explanations for the impact of urban fashion ideals on Black female farmworkers in late nineteenth- and early twentieth-century Texas
- D. To summarize the findings of a study that explored factors influencing a fashion practice among Black female farmworkers in late nineteenth- and early twentieth-century Texas

The following text is from George Marion McClellan's 1895 poem "Eternity."

My spirit swoons, and all my senses cry
For Ocean's breast and covering of the sky.
Rock me to sleep, ye waves, and outward bound,
Just let me drift far out from toil and care,
Where lapping of the waves shall be the sound,
Which mingled with the winds that gently bear
Me on between a peaceful sea and sky,
To make my soothing slumberous lullaby.

Which choice best states the main purpose of the text?

- A. To illustrate the increasing intensity of the speaker's desire to escape ongoing hardship by gliding on the ocean
- B. To contrast the demands of the speaker's everyday life with the serenity of being rocked to sleep by the ocean
- C. To convey the speaker's longing for the ocean to impart a sense of inner tranquility
- D. To justify the speaker's qualms about being transported by the ocean to a quiet destination

Text 1

When the 50-second-long film *Arrival of the Train*—which depicts what its title says, a train pulling into a station—was first shown publicly in 1896, spectators, naïve to the new medium of film and seeing a train appearing to come directly at them, leaped from their seats and fled the room. This moment marks a major cultural shift: a new way of representing and seeing the world had arrived with that train, and nothing would ever be the same.

Text 2

The fact that there is no contemporary evidence that the first audience of *Arrival of the Train* was alarmed has not stopped the story from becoming canonical, even among film historians. But that phenomenon itself is highly revealing. Our belief that the coming of film was transformative is so strong that we invented and keep retelling a founding myth that divides cultural history into a (naïve) “before” and (sophisticated) “after.”

Based on the texts, the author of Text 2 would most likely agree with which statement about the description of the first showing of *Arrival of the Train* in Text 1?

- A. It is more motivated by a perception of the significance of the invention of film than by facts.
- B. It expresses a view about the transformative nature of film that film historians tend to regard as naïve.
- C. It is not based on evidence and reflects film historians’ belief that they are more sophisticated than today’s audiences.
- D. It reflects a misconception that is widely believed because it captures early film audiences’ sense that the new medium was alarming.

Text 1

Fossils of the hominin *Australopithecus africanus* have been found in the Sterkfontein Caves of South Africa, but assigning an age to the fossils is challenging because of the unreliability of dating methods in this context. The geology of Sterkfontein has caused soil layers from different periods to mix, impeding stratigraphic dating, and dates cannot be reliably imputed from those of nearby animal bones since the bones may have been relocated by flooding.

Text 2

Archaeologists used new cosmogenic nuclide dating techniques to reevaluate the ages of *A. africanus* fossils found in the Sterkfontein Caves. This technique involves analyzing the cosmogenic nucleotides in the breccia—the matrix of rock fragments immediately surrounding the fossils. The researchers assert that this approach avoids the potential for misdating associated with assigning ages based on Sterkfontein’s soil layers or animal bones.

Based on the texts, how would the researchers in Text 2 most likely respond to the underlined portion in Text 1?

- A. They would emphasize the fact that the *A. africanus* fossils found in the Sterkfontein Caves may have been corrupted in some way over the years.
- B. They would contend that if analyses of surrounding layers and bones in the Sterkfontein Caves were combined, then the dating of the fossils there would be more accurate.
- C. They would argue that their techniques are better suited than other methods to the unique challenges posed by the Sterkfontein Caves.
- D. They would claim that cosmogenic nuclide dating is reliable in the context of the Sterkfontein Caves because it is applied to the fossils directly.

According to Indian economist and sociologist Radhakamal Mukerjee (1889–1968), the Eurocentric concepts that informed early twentieth-century social scientific methods—for example, the idea that all social relations are reducible to struggles between individuals—had little relevance for India. Making the social sciences more responsive to Indians’ needs, Mukerjee argued, required constructing analytical categories informed by India’s cultural and ecological circumstances. Mukerjee thus proposed the communalist “Indian village” as the ideal model on which to base Indian economic and social policy.

Which choice best describes the overall structure of the text?

- A. The text recounts Mukerjee’s early training in the social scientific disciplines and then lists social policies whose implementation Mukerjee oversaw.
- B. The text mentions some of Mukerjee’s economic theories and then traces their impact on other Indian social scientists of the twentieth century.
- C. The text presents Mukerjee’s critique of the social sciences and then provides an example of his attempts to address issues he identified in his critique.
- D. The text explains an influential economic theory and then demonstrates how that theory was more important to Mukerjee’s work than other social scientists have acknowledged.

Text 1

Soy sauce, made from fermented soybeans, is noted for its umami flavor. Umami—one of the five basic tastes along with sweet, bitter, salty, and sour—was formally classified when its taste receptors were discovered in the 2000s. In 2007, to define the pure umami flavor scientists Rie Ishii and Michael O’Mahony used broths made from shiitake mushrooms and kombu seaweed, and two panels of Japanese and US judges closely agreed on a description of the taste.

Text 2

A 2022 experiment by Manon Jünger et al. led to a greater understanding of soy sauce’s flavor profile. The team initially presented a mixture of compounds with low molecular weights to taste testers who found it was not as salty or bitter as real soy sauce. Further analysis of soy sauce identified proteins, including dipeptides, that enhanced umami flavor and also contributed to saltiness. The team then made a mix of 50 chemical compounds that re-created soy sauce’s flavor.

Based on the texts, if Ishii and O’Mahony (Text 1) and Jünger et al. (Text 2) were aware of the findings of both experiments, they would most likely agree with which statement?

- A. On average, the diets of people in the United States tend to have fewer foods that contain certain dipeptides than the diets of people in Japan have.
- B. Chemical compounds that activate both the umami and salty taste receptors tend to have a higher molecular weight than those that only activate umami taste receptors.
- C. Fermentation introduces proteins responsible for the increase of umami flavor in soy sauce, and those proteins also increase the perception of saltiness.
- D. The broths in the 2007 experiment most likely did not have a substantial amount of the dipeptides that played a key part in the 2022 experiment.

In the 2010s, the price of pop music cassette tapes from the 1980s and 1990s rose dramatically, which had the counterintuitive effect of _____ demand: buyers who hadn't previously wanted to purchase vintage cassettes thronged the market, believing prices would continue to rise and the tapes could be resold later at a profit.

Which choice completes the text with the most logical and precise word or phrase?

- A. exploiting
- B. monetizing
- C. appraising
- D. engendering

In Jane Austen's novel *Mansfield Park*, an almost imperceptible smile from potential suitor Henry Crawford causes the protagonist Fanny Price to blush; her embarrassment grows when she suspects that he is aware of it. This moment—in which Fanny not only infers Henry's mental state through his gestures, but also infers that *he* is drawing inferences about *her* mental state—illustrates what literary scholar George Butte calls “deep intersubjectivity,” a technique for representing interactions between consciousnesses through which Austen's novels derive much of their social and psychological drama.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It states a claim about Austen's skill at representing psychological complexity that is reinforced by an example presented in the following sentence.
- B. It advances an interpretation of an Austen protagonist who is contrasted with protagonists from other Austen novels cited in the following sentence.
- C. It describes a recurring theme in Austen's novels that is the focus of a literary scholar's analysis summarized in the following sentence.
- D. It provides a synopsis of an interaction in an Austen novel that illustrates a literary concept discussed in the following sentence.

Investigating whether shared false visual memories—specific but inaccurate and widely held recollections of images such as product logos—are caused by people’s previous _____ incorrect renditions of the images, researchers Deepasri Prasad and Wilma Bainbridge found that, in fact, such memories are often not explained by familiarity with erroneous versions of the images.

Which choice completes the text with the most logical and precise word or phrase?

- A. compliance with
- B. exposure to
- C. criteria for
- D. forfeiture of

While most animals are incapable of passing somatic mutations—genetic alterations that arise in an organism’s nonreproductive cells—on to their offspring, elkhorn coral (*Acropora palmata*) presents an intriguing ____: in a 2022 study, researchers found that elkhorn coral produced offspring that inherited somatic mutations from a parent.

Which choice completes the text with the most logical and precise word or phrase?

- A. hypothesis
- B. affinity
- C. anomaly
- D. corroboration

Dubautia carrii is a species in a family of plants known collectively as the silversword alliance, all of which grow only on the Hawaiian Islands. Members of this alliance exhibit an extraordinary range of phenotypes, with some species maturing into vines and others into shrubs and trees. All species in the alliance descended from a single ancestral tarweed plant that arrived on the islands around 5 million years ago. The tarweed's descendants diversified into distinct species as they adapted to live in the wide variety of habitats found on the Hawaiian Islands.

Which choice best states the main purpose of the text?

- A. To indicate the specific tarweed ancestor of all plants that grow on the Hawaiian Islands and explain why the plants have such varied physical characteristics
- B. To describe the specific habitat where *Dubautia carrii* are found and identify other plants that share a common ancestor with them
- C. To describe the silversword alliance and explain how the plant family became so varied
- D. To advance the claim that all plants on the Hawaiian Islands are part of the silversword alliance and list possible ancestors of the plants that make up the alliance

Economist Marco Castillo and colleagues showed that nuisance costs—the time and effort people must spend to make donations—reduce charitable giving. Charities can mitigate this effect by compensating donors for nuisance costs, but those costs, though variable, are largely _____ donation size, so charities that compensate donors will likely favor attracting a few large donors over many small donors.

Which choice completes the text with the most logical and precise word or phrase?

- A. supplemental to
- B. predictive of
- C. independent of
- D. subsumed in

While scholars believe many Mesoamerican cities influenced each other, direct evidence of such influence is difficult to ascertain. However, recent excavations in a sector of Tikal (Guatemala) unearthed a citadel that shows _____ Teotihuacán (Mexico) architecture—including a near replica of a famed Teotihuacán temple—providing tangible evidence of outside influence in portions of Tikal.

Which choice completes the text with the most logical and precise word or phrase?

- A. refinements of
- B. precursors of
- C. commonalities with
- D. animosities toward

During Rome's republican period, which ended in the first century BCE, libraries were predominantly owned by wealthy individuals who tightly controlled access to their book collections. The first public library became available in Rome in 28 BCE and was soon followed by one commissioned by Emperor Augustus. As modern scholar Fabio Fernandes notes, however, these two traditions aren't as distinct as they seem, as both the emperor and the private library owners viewed their libraries as extensions of their personal patronage, just on vastly differing scales.

Which choice best states the main purpose of the text?

- A. To give a brief overview of public access to libraries throughout Rome's republican period
- B. To contend that early imperial leaders in Rome wielded too much influence over libraries
- C. To assert that private and early public libraries in ancient Rome had an essential similarity
- D. To call into question the notion that private Roman libraries disappeared during the first century BCE

Text corpora such as the Bank of English are enormous collections of electronically stored texts that can be used for empirical testing of hypotheses regarding the frequency of typical word usage. If one has a _____ that the word “get” has a high incidence in English, for example, an analysis of a corpus can support that hypothesis by showing that “get” is the fifth most commonly used verb.

Which choice completes the text with the most logical and precise word or phrase?

- A. recognition
- B. scheme
- C. supposition
- D. synopsis

As an undergraduate researcher in anthropology, Jennifer C. Chen contributed to a groundbreaking study challenging the accepted view that among prehistoric peoples, female participation in hunting was _____. The research team's review of data from late Pleistocene and early Holocene burials in the Americas revealed that, in fact, as many as half of the hunters in those populations were female.

Which choice completes the text with the most logical and precise word or phrase?

- A. inevitable
- B. satisfactory
- C. negligible
- D. commonplace

Text 1

Virginia Woolf's 1928 novel *Orlando* is an oddity within her body of work. Her other major novels consist mainly of scenes of everyday life and describe their characters' interior states in great detail, whereas *Orlando* propels itself through a series of fantastical events and considers its characters' psychology more superficially. Woolf herself sometimes regarded the novel as a minor work, even admitting once that she "began it as a joke."

Text 2

Like Woolf's other great novels, *Orlando* portrays how people's memories inform their experience of the present. Like those works, it examines how people navigate social interactions shaped by gender and social class. Though it is lighter in tone—more entertaining, even—this literary "joke" nonetheless engages seriously with the themes that motivated the four or five other novels by Woolf that have achieved the status of literary classics.

Based on the texts, how would the author of Text 2 most likely respond to the assessment of *Orlando* presented in Text 1?

- A. By conceding that Woolf's talents were best suited to serious novels but asserting that the humor in *Orlando* is often effective
- B. By agreeing that *Orlando* is less impressive than certain other novels by Woolf but arguing that it should still be regarded as a classic
- C. By acknowledging that *Orlando* clearly differs from Woolf's other major novels but insisting on its centrality to her body of work nonetheless
- D. By concurring that the reputation of *Orlando* as a minor work has led readers to overlook this novel but maintaining that the reputation is unearned

Text 1

Like the work of Ralph Ellison before her, Toni Morrison's novels feature scenes in which characters deliver sermons of such length and verbal dexterity that for a time, the text exchanges the formal parameters of fiction for those of oral literature. Given the many other echoes of Ellison in Morrison's novels, both in structure and prose style, these scenes suggest Ellison's direct influence on Morrison.

Text 2

In their destabilizing effect on literary form, the sermons in Morrison's works recall those in Ellison's. Yet literature by Black Americans abounds in moments where interpolated speech erodes the division between oral and written forms that literature in English has traditionally observed. Morrison's use of the sermon is attributable not only to the influence of Ellison but also to a community-wide strategy of resistance to externally imposed literary conventions.

Based on the texts, how would the author of Text 2 most likely characterize the underlined claim in Text 1?

- A. As failing to consider Ellison's and Morrison's equivalent uses of the sermon within the wider cultural context in which they wrote
- B. As misunderstanding the function of sermons in novels by Black American writers other than Ellison and Morrison
- C. As disregarding points of structural and stylistic divergence between the works of Ellison and those of Morrison
- D. As being indebted to the tradition of resisting literary conventions that privilege written forms, such as novels, over sermons and other oral forms

Despite how _____ they may seem on first glance, the works of the philosopher Georg Hegel have proven to be quite influential. Readers who manage to penetrate Hegel's frustratingly dense prose often find that his thrillingly original and profound ideas wind up shaping their own thinking.

Which choice completes the text with the most logical and precise word or phrase?

- A. authoritative
- B. opaque
- C. empirical
- D. facile

Vadamalai Elangovan and Ganapathi Marimuthu showed that high moonlight intensity inhibits the activity of the greater short-nosed fruit bat (*Cynopterus sphinx*), a result explicable in terms of benefits and costs: greater lunar intensity may not enable the bats to increase foraging success enough to offset the higher chance of detection by predatory owls or hawks. Most other nocturnal mammals respond to lunar intensity variations similarly to greater short-nosed fruit bats, but mongoose lemurs (*Eulemur mongoz*) display the opposite pattern, as their heavy reliance on visual foraging results in a different balance of reward and risk.

Which choice best describes the overall structure of the text?

- A. It discusses two different responses to the same natural phenomenon, explains why one of those responses occurs, and then suggests that the other response still needs to be explained.
- B. It describes and accounts for a finding, characterizes the finding as representative of a general pattern, and then describes and accounts for an exception to that pattern.
- C. It presents and explains a study result, indicates that the result is similar to the results of many other studies, and then attributes a conflicting study result to a difference in that study's methods.
- D. It introduces an observation of a behavioral pattern, presents an explanation for the pattern, and then describes an exception to the pattern that casts doubt on that explanation.

The 1967 release of Harold Cruse's book *The Crisis of the Negro Intellectual* isolated him from almost all other scholars and activists of the American Civil Rights Movement—though many of those thinkers disagreed with each other, he nonetheless found ways to disagree with them all. He thought that activists who believed that Black people such as himself should culturally assimilate were naïve. But he also sharply criticized Black nationalists such as Marcus Garvey who wanted to establish independent, self-contained Black economies and societies, even though Cruse himself identified as a Black nationalist.

Which choice best describes the function of the underlined sentence in the text as a whole?

- A. It describes a direction that Cruse felt the Civil Rights Movement ought to take.
- B. It indicates that Cruse's reputation as a persistent antagonist of other scholars is undeserved.
- C. It describes a controversy that Cruse's work caused within the Black nationalist movement.
- D. It helps explain Cruse's position with respect to the community of civil rights thinkers.

The following text is adapted from Herman Melville's 1857 novel *The Confidence-Man*. Humphry Davy was a prominent British chemist and inventor.

Years ago, a grave American savant, being in London, observed at an evening party there, a certain coxcombical fellow, as he thought, an absurd ribbon in his lapel, and full of smart [banter], whisking about to the admiration of as many as were disposed to admire. Great was the savant's disdain; but, chancing ere long to find himself in a corner with the jackanapes, got into conversation with him, when he was somewhat ill-prepared for the good sense of the jackanapes, but was altogether thrown aback, upon subsequently being [informed that he was] no less a personage than Sir Humphry Davy.

Which choice best states the main purpose of the text?

- A. It portrays the thoughts of a character who is embarrassed about his own behavior.
- B. It presents an account of a misunderstanding.
- C. It offers a short history of how a person came to be famous.
- D. It explains why one character dislikes another.

Despite stated claims of global relevance, much major research on income inequality performed in the 2010s suffered from a myopic focus on a few countries in North America and Western Europe, partly due to limited data availability. Researchers would later _____ this shortcoming after gaining new access to banking records located in nations in Africa, such as Tunisia, and Eastern Europe, such as Hungary.

Which choice completes the text with the most logical and precise word or phrase?

- A. presuppose
- B. categorize
- C. mitigate
- D. validate

Text 1

In 2007, a team led by Alice Storey analyzed a chicken bone found in El Arenal, Chile, dating it to 1321–1407 CE—over a century before Europeans invaded the region, bringing their own chickens. Storey also found that the El Arenal chicken shared a unique genetic mutation with the ancient chicken breeds of the Polynesian Islands in the Pacific. Thus, Polynesian peoples, not later Europeans, probably first introduced chickens to South America.

Text 2

An Australian research team weakened the case for a Polynesian origin for the El Arenal chicken by confirming that the mutation identified by Storey has occurred in breeds from around the world. More recently, though, a team led by Augusto Luzuriaga-Neira found that South American chicken breeds and Polynesian breeds share other genetic markers that European breeds lack. Thus, the preponderance of evidence now favors a Polynesian origin.

Based on the texts, how would the author of Text 2 most likely respond to the underlined claim in Text 1?

- A. By broadly agreeing with the claim but objecting that the timeline it presupposes conflicts with the findings of the genetic analysis conducted by Storey's team
- B. By faulting the claim for implying that domestic animals couldn't have been transferred from South America to the Polynesian Islands as well
- C. By critiquing the claim for being based on an assumption that before the European invasion of South America, the chickens of Europe were genetically uniform
- D. By noting that while the claim is persuasive, the findings of Luzuriaga-Neira's team provide stronger evidence for it than the findings of the genetic analysis conducted by Storey do

Some studies have suggested that posture can influence cognition, but we should not overstate this phenomenon. A case in point: In a 2014 study, Megan O'Brien and Alaa Ahmed had subjects stand or sit while making risky simulated economic decisions. Standing is more physically unstable and cognitively demanding than sitting; accordingly, O'Brien and Ahmed hypothesized that standing subjects would display more risk aversion during the decision-making tasks than sitting subjects did, since they would want to avoid further feelings of discomfort and complicated risk evaluations. But O'Brien and Ahmed actually found no difference in the groups' performance.

Which choice best states the main purpose of the text?

- A. It argues that research findings about the effects of posture on cognition are often misunderstood, as in the case of O'Brien and Ahmed's study.
- B. It presents the study by O'Brien and Ahmed to critique the methods and results reported in previous studies of the effects of posture on cognition.
- C. It explains a significant problem in the emerging understanding of posture's effects on cognition and how O'Brien and Ahmed tried to solve that problem.
- D. It discusses the study by O'Brien and Ahmed to illustrate why caution is needed when making claims about the effects of posture on cognition.

Raymond Antrobus, an accomplished poet and writer of prose, recently released his debut spoken word poetry album, *The First Time I Wore Hearing Aids*, in collaboration with producer Ian Brennan. The album contains both autobiographical and reflective pieces combining Antrobus's spoken words with Brennan's fragmented audio elements and pieces of music to convey how people who are deaf may experience sound, both its presence and absence. Some critics suggest that the album questions the function of sound in the world, highlighting that the experience of sound is multifaceted.

Which choice best describes the overall structure of the text?

- A. It introduces a collaborative spoken word poetry project, details the approach taken to produce the work, and then provides an example of critique the album received upon release.
- B. It mentions a collection of spoken word poems, distinguishes one poem as being an exemplar on the album, and then offers a summary of the subject matter of the whole collection.
- C. It summarizes the efforts to produce a collection of spoken word poems, presents biographies of two people who worked on the album, and speculates about the meaning behind the poetry.
- D. It connects two artists to the same spoken word poetry project, explains the extent of their collaboration on each poem, and then provides an overview of the technique used to produce the work.

In a study of residential mobility in Washington, DC, LaTanya Brown-Robertson et al. found that increases in household spending power (via tax credits) had mixed effects on mobility decisions, possibly because residential location choices are not fully reducible to economic factors but are also _____ noneconomic factors such as access to strong local social networks.

Which choice completes the text with the most logical and precise word or phrase?

- A. transposed with
- B. mediated by
- C. decoupled from
- D. antecedent to

According to a US tax policy expert, state taxes are _____ other factors when considering an interstate move. Even significant differences in state taxation have almost no effect on most people's decisions, while differences in employment opportunities, housing availability, and climate are strong influences.

Which choice completes the text with the most logical and precise word or phrase?

- A. consistent with
- B. representative of
- C. overshadowed by
- D. irrelevant to

Close analysis of the painting *Girl with a Flute*, long attributed to the seventeenth-century Dutch painter Johannes Vermeer, has revealed subtle deviations from the artist's signature techniques. These variations suggest that the work may be that of a student under Vermeer's tutelage—potentially _____ our understanding of Vermeer as a solitary artist.

Which choice completes the text with the most logical and precise word or phrase?

- A. negating
- B. prefiguring
- C. entrenching
- D. substantiating

Text 1

Africa's Sahara region—once a lush ecosystem—began to dry out about 8,000 years ago. A change in Earth's orbit that affected climate has been posited as a cause of desertification, but archaeologist David Wright also attributes the shift to Neolithic peoples. He cites their adoption of pastoralism as a factor in the region drying out: the pastoralists' livestock depleted vegetation, prompting the events that created the Sahara Desert.

Text 2

Research by Chris Brierley et al. challenges the idea that Neolithic peoples contributed to the Sahara's desertification. Using a climate-vegetation model, the team concluded that the end of the region's humid period occurred 500 years earlier than previously assumed. The timing suggests that Neolithic peoples didn't exacerbate aridity in the region but, in fact, may have helped delay environmental changes with practices (e.g., selective grazing) that preserved vegetation.

Based on the texts, how would Chris Brierley (Text 2) most likely respond to the discussion in Text 1?

- A. By pointing out that given the revised timeline for the end of the Sahara's humid period, the Neolithic peoples' mode of subsistence likely didn't cause the region's desertification
- B. By claiming that pastoralism was only one of many behaviors the Neolithic peoples took part in that may have contributed to the Sahara's changing climate
- C. By insisting that pastoralism can have both beneficial and deleterious effects on a region's vegetation and climate
- D. By asserting that more research needs to be conducted into factors that likely contributed to the desertification of the Sahara region

The following text is adapted from Oscar Wilde's 1895 play *The Importance of Being Earnest*.

CECILY: Have we got to part?

ALGERNON: I am afraid so. It's a very painful parting.

CECILY: It is always painful to part from people whom one has known for a very brief space of time. The absence of old friends one can endure with equanimity. But even a momentary separation from anyone to whom one has just been introduced is almost unbearable.

As used in the text, what does the word "endure" most nearly mean?

- A. Regret
- B. Persist
- C. Tolerate
- D. Encourage

The following text is from John Dryden's 1697 translation of Virgil's poem the *Aeneid*, written in the first century BCE. Queen Dido has just heard the Trojan hero Aeneas recount a sad tale.

But anxious cares already seiz'd the queen:
She fed within her veins a flame unseen;
The hero's valor, acts, and birth inspire
Her soul with love, and fan the secret fire.
His words, his looks, imprinted in her heart,
Improve the passion, and increase the smart.

Which choice best states the main purpose of the text?

- A. To indicate that Dido is hiding her skepticism of Aeneas's account
- B. To emphasize that Dido has been deeply affected by Aeneas's story
- C. To imply that Dido engages both emotionally and intellectually with Aeneas
- D. To suggest that Dido is worried that Aeneas might discover a secret she is keeping from him

Text 1

In 1916, H. Dugdale Sykes disputed claims that *The Two Noble Kinsmen* was coauthored by William Shakespeare and John Fletcher. Sykes felt Fletcher's contributions to the play were obvious—Fletcher had a distinct style in his other plays, so much so that lines with that style were considered sufficient evidence of Fletcher's authorship. But for the lines not deemed to be by Fletcher, Sykes felt that their depiction of women indicated that their author was not Shakespeare but Philip Massinger.

Text 2

Scholars have accepted *The Two Noble Kinsmen* as coauthored by Shakespeare since the 1970s: it appears in all major one-volume editions of Shakespeare's complete works. Though scholars disagree about who wrote what exactly, it is generally held that on the basis of style, Shakespeare wrote all of the first act and most of the last, while John Fletcher authored most of the three middle acts.

Based on the texts, both Sykes in Text 1 and the scholars in Text 2 would most likely agree with which statement?

- A. John Fletcher's writing has a unique, readily identifiable style.
- B. The women characters in John Fletcher's plays are similar to the women characters in Philip Massinger's plays.
- C. *The Two Noble Kinsmen* belongs in one-volume compilations of Shakespeare's complete plays.
- D. Philip Massinger's style in the first and last acts of *The Two Noble Kinsmen* is an homage to Shakespeare's style.

Some foraging models predict that the distance bees travel when foraging will decline as floral density increases, but biologists Shalene Jha and Claire Kremen showed that bees' behavior is inconsistent with this prediction if flowers in dense patches are _____. bees will forage beyond patches of low species richness to acquire multiple resource types.

Which choice completes the text with the most logical and precise word or phrase?

- A. depleted
- B. homogeneous
- C. immature
- D. dispersed

Rejecting the premise that the literary magazine *Ebony and Topaz* (1927) should present a unified vision of Black American identity, editor Charles S. Johnson fostered his contributors' diverse perspectives by promoting their authorial autonomy. Johnson's self-effacement diverged from the editorial stances of W.E.B. Du Bois and Alain Locke, whose decisions for their publications were more _____.

Which choice completes the text with the most logical and precise word or phrase?

- A. proficient
- B. dogmatic
- C. ambiguous
- D. unpretentious

Mathematician Claude Shannon is widely regarded as a foundational figure in information theory. His most important paper, “A Mathematical Theory of Communication,” published in 1948 when he was employed at Bell Labs, utilized a concept called a “binary digit” (shortened to “bit”) to measure the amount of information in any signal and determine the fastest rate at which information could be transmitted while still being reliably decipherable. Robert Gallager, one of Shannon’s colleagues, said that the bit was “[Shannon’s] discovery, and from it the whole communications revolution has sprung.”

Which choice best describes the overall structure of the text?

- A. It presents a theoretical concept, illustrates how the name of the concept has changed, and shows how the name has entered common usage.
- B. It introduces a respected researcher, describes an aspect of his work, and suggests why the work is historically significant.
- C. It names the company where an important mathematician worked, details the mathematician’s career at the company, and provides an example of the recognition he received there.
- D. It mentions a paper, offers a summary of the paper’s findings, and presents a researcher’s commentary on the paper.

Text 1

What factors influence the abundance of species in a given ecological community? Some theorists have argued that historical diversity is a major driver of how diverse an ecological community eventually becomes: differences in community diversity across otherwise similar habitats, in this view, are strongly affected by the number of species living in those habitats at earlier times.

Text 2

In 2010, a group of researchers including biologist Carla Cáceres created artificial pools in a New York forest. They stocked some pools with a diverse mix of zooplankton species and others with a single zooplankton species and allowed the pool communities to develop naturally thereafter. Over the course of four years, Cáceres and colleagues periodically measured the species diversity of the pools, finding—contrary to their expectations—that by the end of the study there was little to no difference in the pools' species diversity.

Based on the texts, how would Cáceres and colleagues (Text 2) most likely describe the view of the theorists presented in Text 1?

- A. It is largely correct, but it requires a minor refinement in light of the research team's results.
- B. It is not compelling as a theory regardless of any experimental data collected by the research team.
- C. It may seem plausible, but it is not supported by the research team's findings.
- D. It probably holds true only in conditions like those in the research team's study.

Text 1

The fossil record suggests that mammoths went extinct around 11 thousand years (kyr) ago. In a 2021 study of environmental DNA (eDNA)—genetic material shed into the environment by organisms—in the Arctic, Yucheng Wang and colleagues found mammoth eDNA in sedimentary layers formed millennia later, around 4 kyr ago. To account for this discrepancy, Joshua H. Miller and Carl Simpson proposed that arctic temperatures could preserve a mammoth carcass on the surface, allowing it to leach DNA into the environment, for several thousand years.

Text 2

Wang and colleagues concede that eDNA contains DNA from both living organisms and carcasses, but for DNA to leach from remains over several millennia requires that the remains be perpetually on the surface. Scavengers and weathering in the Arctic, however, are likely to break down surface remains well before a thousand years have passed.

Which choice best describes how Text 1 and Text 2 relate to each other?

- A. Text 1 discusses two approaches to studying mammoth extinction without advocating for either, whereas Text 2 advocates for one approach over the other.
- B. Text 1 presents findings by Wang and colleagues and gives another research team's attempt to explain those findings, whereas Text 2 provides additional detail that calls that explanation into question.
- C. Text 1 describes Wang and colleagues' study and a critique of their methodology, whereas Text 2 offers additional details showing that methodology to be sound.
- D. Text 1 argues that new research has undermined the standard view of when mammoths went extinct, whereas Text 2 suggests a way to reconcile the standard view with that new research.

Text 1

Despite its beautiful prose, *The Guns of August*, Barbara Tuchman's 1962 analysis of the start of World War I, has certain weaknesses as a work of history. It fails to address events in Eastern Europe just before the outbreak of hostilities, thereby giving the impression that Germany was the war's principal instigator. Had Tuchman consulted secondary works available to her by scholars such as Luigi Albertini, she would not have neglected the influence of events in Eastern Europe on Germany's actions.

Text 2

Barbara Tuchman's *The Guns of August* is an engrossing if dated introduction to World War I. Tuchman's analysis of primary documents is laudable, but her main thesis that European powers committed themselves to a catastrophic outcome by refusing to deviate from military plans developed prior to the conflict is implausibly reductive.

Which choice best describes a difference in how the authors of Text 1 and Text 2 view Barbara Tuchman's *The Guns of August*?

- A. The author of Text 1 argues that Tuchman should have relied more on the work of other historians, while the author of Text 2 implies that Tuchman's most interesting claims result from her original research.
- B. The author of Text 1 believes that the scope of Tuchman's research led her to an incorrect interpretation, while the author of Text 2 believes that Tuchman's central argument is overly simplistic.
- C. The author of Text 1 asserts that the writing style of *The Guns of August* makes it worthwhile to read despite any perceived deficiency in Tuchman's research, while the author of Text 2 focuses exclusively on the weakness of Tuchman's interpretation of events.
- D. The author of Text 1 claims that Tuchman would agree that World War I was largely due to events in Eastern Europe, while the author of Text 2 maintains that Tuchman would say that Eastern European leaders were not committed to military plans in the same way that other leaders were.